ENAE464 - 0202

Lab01: Pressure Measurements of Inviscid Flow over Cylinder

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Dr. Silbaugh, 02:00 PM

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Enclosed is the technical report for the *Flow Over Cylinder* laboratory experiment. This report presents the experimental methodology, results, and analysis of flow characteristics observed around a circular cylinder, including pressure distribution, drag coefficient determination, and comparison with theoretical predictions. Please feel free to contact us with any questions regarding the contents of this report.

Respectfully,
Mikołaj Kostrzewa & Vai Srivastava

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1 Abstract

This report presents the results of an experimental investigation of the pressure distribution over a circular cylinder of diameter 1.27 cm in a vertical wind tunnel. Surface pressure measurements were obtained by rotating a single pressure tap in $5\,°$ increments from $0\,°$ to $180\,°$, and the freestream velocity was determined from pitot-static tube readings via Bernoulli's equation. The measured pressure coefficient distribution was compared with the inviscid potential flow prediction $C_p = 1 - 4\sin^2\theta$, and the drag coefficient was computed by trapezoidal integration of the surface pressure. The experimental data agreed closely with inviscid theory over the forward portion of the cylinder ($0\,° \leq \theta \leq 30\,°$), but diverged significantly beyond $\theta \approx 30\,°$, where boundary layer separation caused the pressure coefficient to plateau near $C_p \approx -1$ rather than reaching the theoretical minimum of -3. The resulting asymmetry between fore and aft pressure yielded a nonzero drag coefficient, in contrast to the zero-drag prediction of inviscid theory (d'Alembert's paradox).

2 Introduction

For the inviscid-flow model used in this lab, we assume a potential-flow field that is:

- **Incompressible (continuity):** $\nabla \cdot \vec{\mathbf{V}} = 0$.
- **Irrotational:** $\nabla \times \vec{\mathbf{V}} = 0$.

If $\nabla \times \vec{\mathbf{V}} = 0$, then a *velocity potential* ϕ exists such that $\vec{\mathbf{V}} = \nabla\phi$. Substituting into continuity gives Laplace's equation:

$$\nabla^2\phi = 0 \qquad \left(\text{i.e., } \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} = 0\right). \tag{2.0.1}$$

In order to model flow over a 2D cylinder, we use an approach presented in [1], which uses a superposition of a uniform flow and a doublet flow, as presented in the figure below [1, Fig. 3.26]:

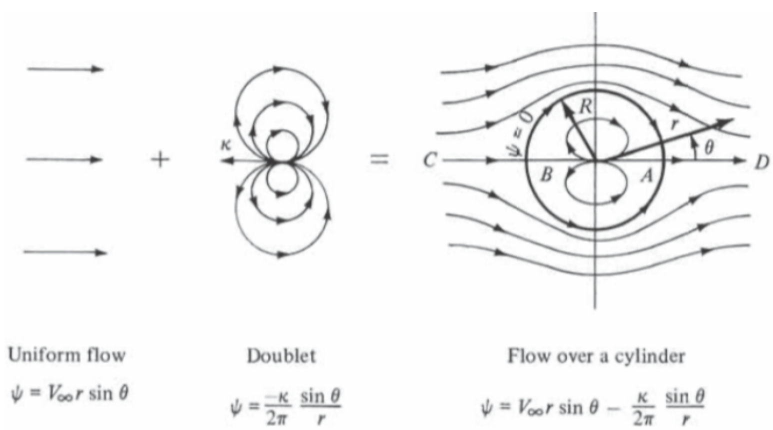

Figure 2.0.1: Superposition of uniform flow and a doublet to model nonlifting potential flow over a circular cylinder.

In polar coordinates (r, θ), the stream function and velocity potential for both the uniform flow and the doublet are given by:

$$\phi_U = V_\infty r\cos(\theta) \qquad \text{(velocity potential: uniform flow)} \tag{2.0.2}$$

$$\psi_U = V_\infty r\sin(\theta) \qquad \text{(stream function: uniform flow)} \tag{2.0.3}$$

$$\phi_D = \frac{\kappa}{2\pi}\frac{\cos(\theta)}{r} \qquad \text{(velocity potential: doublet)} \tag{2.0.4}$$

$$\psi_D = -\frac{\kappa}{2\pi}\frac{\sin(\theta)}{r} \qquad \text{(stream function: doublet)} \tag{2.0.5}$$

where V_∞ is the freestream speed and κ is the doublet strength.
Superposing the uniform flow and doublet stream functions gives

$$\begin{aligned}\psi &= \psi_U + \psi_D && (2.0.6)\\ &= V_\infty r\sin(\theta) - \frac{\kappa}{2\pi}\frac{\sin(\theta)}{r} && (2.0.7)\\ &= V_\infty r\sin(\theta)\left(1 - \frac{R^2}{r^2}\right), && (2.0.8)\end{aligned}$$

where the cylinder radius is defined by

$$R^2 \equiv \frac{\kappa}{2\pi V_\infty}. \tag{2.0.9}$$

The corresponding velocity field in polar coordinates follows from the stream function relations $V_r = \frac{1}{r}\frac{\partial\psi}{\partial\theta}$ and $V_\theta = -\frac{\partial\psi}{\partial r}$:

$$\begin{aligned}V_r &= \frac{1}{r}\frac{\partial\psi}{\partial\theta} = \frac{1}{r}\left(V_\infty r\cos(\theta)\right)\left(1 - \frac{R^2}{r^2}\right) && (2.0.10)\\ &= \left(1 - \frac{R^2}{r^2}\right)V_\infty\cos(\theta), && (2.0.11)\\ V_\theta &= -\frac{\partial\psi}{\partial r} && (2.0.12)\\ &= -\left(1 + \frac{R^2}{r^2}\right)V_\infty\sin(\theta). && (2.0.13)\end{aligned}$$

At the surface of the cylinder, $r = R$, the velocity field is:

$$\begin{aligned}V_r &= \left(1 - \frac{R^2}{R^2}\right)V_\infty\cos(\theta) = 0, && (2.0.14)\\ V_\theta &= -\left(1 + \frac{R^2}{R^2}\right)V_\infty\sin(\theta) = -2V_\infty\sin(\theta). && (2.0.15)\end{aligned}$$

This aligns with the assumption that the velocity normal to the stationary surface must be zero, and due to polar coordinates, the velocity normal to the surface is given by V_r. Since V_θ is perpendicular to the surface, due to construction of the coordinate system, it's equivalent to a tangential velocity, and since normal velocity is zero, $V = V_\theta = V_\infty\sin(\theta)$.
The pressure coefficient is given by:

$$\begin{aligned}C_p &= 1 - \frac{V^2}{V_\infty^2} && (2.0.16)\\ V &= -2V_\infty\sin(\theta) && (2.0.17)\\ \frac{V^2}{V_\infty^2} &= 4\sin^2\theta && (2.0.18)\end{aligned}$$

$$\boxed{C_p = 1 - \frac{V^2}{V_\infty^2} = 1 - 4\sin^2\theta} \tag{2.0.19}$$

3 Experimental Apparatus and Procedures

3.1 Experimental Apparatus

The experimental apparatus for this lab consisted of a wind tunnel whose test chamber contained a small metal cylinder, with diameter 1.27 cm. This wind tunnel was oriented such that the flow of air through the test chamber happened from top to bottom. Being above the direction of flow, the air reservoir was set up to hold unpressurized air (being left only at ambient room pressure). From there, the air is accelerated through the test chamber, and exits out the back into the room. Installed in the test chamber is the cylinder, connected to a scale with demarcations at every $5\,^{\circ}$. Installed within the wind tunnel were 3 pitot-static tubes connected to a manometer: tracking the pressure in the reservoir P_0, upstream of the cylinder P_∞, and on the surface of the cylinder P—aligned with the $0\,^{\circ}$ angle.

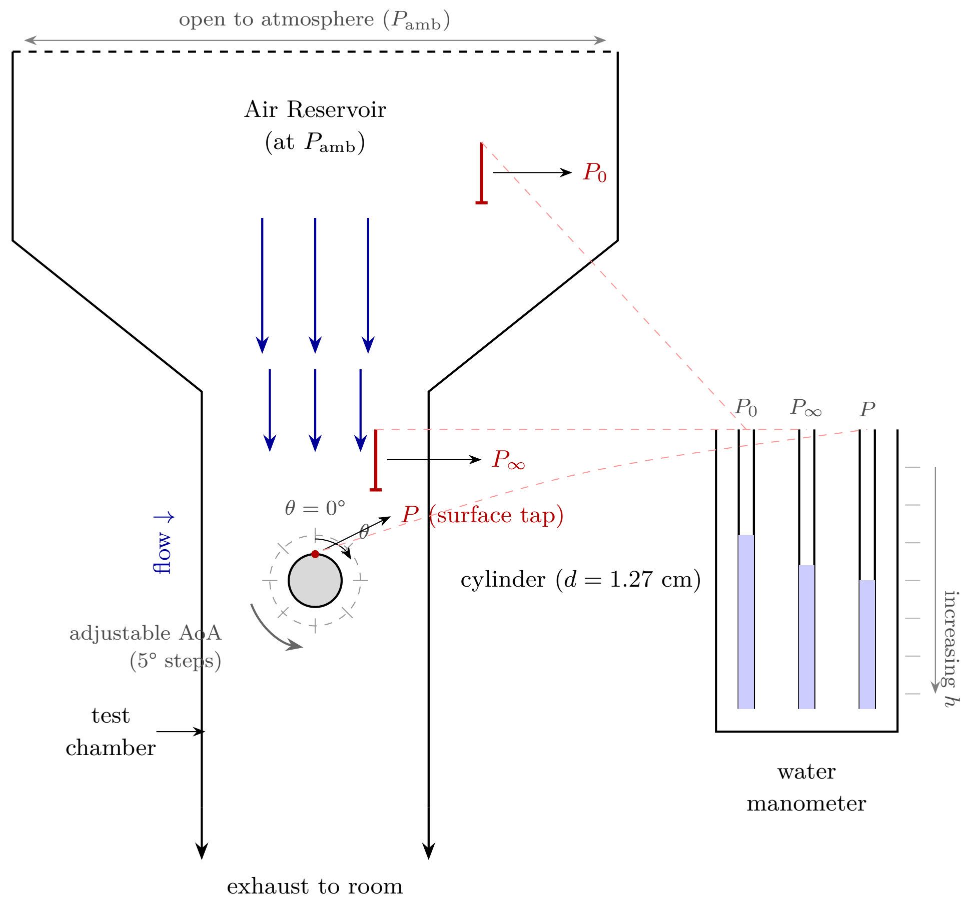

Figure 3.1.1: Wind Tunnel Apparatus

3.2 Operating Technique

With the wind tunnel operating at a fixed flow speed, the surface pressure distribution was measured by rotating the cylinder from $0\,^{\circ}$ to $180\,^{\circ}$ in $5\,^{\circ}$ steps. At each angular position, the water manometer readings for the reservoir pressure P_0, the freestream static pressure P_∞, and the surface tap pressure P were recorded.

4 Results

4.1 Operating Conditions

Table 4.1.1: Ambient Room Conditions during Experiment

Quantity	Value	Uncertainty
Room Temperature, T	$25\,°\mathrm{C}$	$\pm 0.5\,°\mathrm{C}$
Room Pressure, P	$1009.0\,\mathrm{hPa}$	$\pm 0.05\,\mathrm{hPa}$

The density of air (ρ) can be calculated using the Ideal Gas Law:

$$\rho = \frac{P}{RT} \tag{4.1.1}$$

where:

- P is the absolute pressure (in SI units: Pascals Pa)
- T is the absolute temperature (in Kelvin K)
- R is the specific gas constant for dry air, $R = 287.05\,\mathrm{J\,kg^{-1}\,K^{-1}}$

Given measurements:

$$P = 1009.0\,\mathrm{hPa} = 100\,900\,\mathrm{Pa} \tag{4.1.2}$$

$$T = 25\,\mathrm{C} = 25 + 273.15 = 298.15\,\mathrm{K} \tag{4.1.3}$$

$$R = 287.05\,\frac{\mathrm{J}}{\mathrm{kg\,K}} \tag{4.1.4}$$

$$\rho = \frac{100\,900\,\mathrm{Pa}}{287.05\,\mathrm{J\,kg^{-1}\,K^{-1}}} \times 298.15\,\mathrm{K} \tag{4.1.5}$$

Calculate:

$$\rho = \frac{100900}{287.05 \times 298.15} \tag{4.1.6}$$

$$= \frac{100900}{85542.5} \tag{4.1.7}$$

$$\approx 1.18\,\frac{\mathrm{kg}}{\mathrm{m^3}} \tag{4.1.8}$$

Thus, **the ambient air density during the experiment is**:

$$\boxed{\rho = 1.18\,\frac{\mathrm{kg}}{\mathrm{m^3}}} \tag{4.1.9}$$

4.2 Measurements

Table 4.2.1: Measured Relative Presssure, Pressure Coefficient, and Pressure vs. Angle of Attack

`./outputs/text/pressure_vs_theta.csv`

	θ [°]	$P - P_\infty$ [Pa]	C_P	P [Pa]
01	0.0	744.069	1.0000	101644.069
02	5.0	714.698	0.9481	101614.698
03	10.0	675.536	0.8961	101575.536
04	15.0	577.632	0.7564	101477.632
05	20.0	450.357	0.5897	101350.357
06	25.0	313.292	0.4156	101213.292
07	30.0	146.856	0.1923	101046.856
08	35.0	-39.162	-0.0519	100860.838
09	40.0	-225.179	-0.2987	100674.821
10	45.0	-411.196	-0.5455	100488.804
11	50.0	-577.632	-0.7564	100322.368
12	55.0	-675.536	-0.8846	100224.464
13	60.0	-783.230	-1.0390	100116.770
14	65.0	-822.392	-1.0909	100077.608
15	70.0	-832.182	-1.0897	100067.818
16	75.0	-763.650	-1.0000	100136.350
17	80.0	-744.069	-0.9744	100155.931
18	85.0	-724.488	-0.9610	100175.512
19	90.0	-714.698	-0.9481	100185.302
20	95.0	-695.117	-0.9221	100204.883
21	100.0	-695.117	-0.9103	100204.883
22	105.0	-704.907	-0.9351	100195.093
23	110.0	-714.698	-0.9481	100185.302
24	115.0	-714.698	-0.9481	100185.302
25	120.0	-714.698	-0.9481	100185.302
26	125.0	-734.279	-0.9740	100165.721
27	130.0	-744.069	-0.9870	100155.931
28	135.0	-744.069	-0.9870	100155.931
29	140.0	-753.859	-1.0000	100146.141
30	145.0	-763.650	-1.0130	100136.350
31	150.0	-753.859	-1.0000	100146.141
32	155.0	-744.069	-0.9870	100155.931
33	160.0	-724.488	-0.9737	100175.512
34	165.0	-724.488	-0.9610	100175.512
35	170.0	-704.907	-0.9351	100195.093
36	175.0	-704.907	-0.9351	100195.093
37	180.0	-704.907	-0.9351	100195.093

4.3 Quantities of Interest

4.3.1 Mean Wind Tunnel Velocity

The mean wind tunnel velocity, U_∞—being the same as the freestream velocity—can be obtained directly from the pitot-static pressure readings. The pitot-static tube provides two pressure readings via the water manometer: a stagnation (total) pressure tap reading h_0 and a freestream static pressure tap reading h_∞. Thus, the dynamic pressure is

$$q_\infty = P_0 - P_\infty = \rho_w \, g \, (h_0 - h_\infty) \, , \tag{4.3.1}$$

where ρ_w is the density of water and g is gravitational acceleration. Applying Bernoulli's equation along a streamline between the freestream and the stagnation point,

$$P_0 = P_\infty + \frac{1}{2} \rho_\infty \, U_\infty^2 . \tag{4.3.2}$$

We identify $q_\infty = \frac{1}{2} \rho_\infty \, U_\infty^2$ and solve for the freestream velocity:

$$\boxed{U_\infty = \sqrt{\frac{2 \, q_\infty}{\rho_\infty}} = \sqrt{\frac{2 \, \rho_w \, g \, (h_0 - h_\infty)}{\rho_\infty}}} \tag{4.3.3}$$

From the values obtained in the measurements, **the mean wind tunnel velocity is**:

$$\boxed{U_\infty = 35.79 \, \frac{\mathrm{m}}{\mathrm{s}}} \tag{4.3.4}$$

4.3.2 Coefficient of Pressure

The pressure coefficient C_p non-dimensionalises the surface pressure relative to the freestream by the dynamic pressure:

$$C_p = \frac{P - P_\infty}{\frac{1}{2} \rho_\infty \, U_\infty^2} = \frac{P - P_\infty}{q_\infty} . \tag{4.3.5}$$

Substituting the manometer-based expressions,

$$C_p = \frac{\rho_w \, g \, (h_P - h_\infty)}{\rho_w \, g \, (h_0 - h_\infty)} = \frac{h_P - h_\infty}{h_0 - h_\infty} . \tag{4.3.6}$$

Note that the water density and gravitational acceleration cancel, so the pressure coefficient reduces to a simple ratio of manometer height differences. For the theoretical comparison, we use (2.0.19) as the baseline for comparison with experimental data.

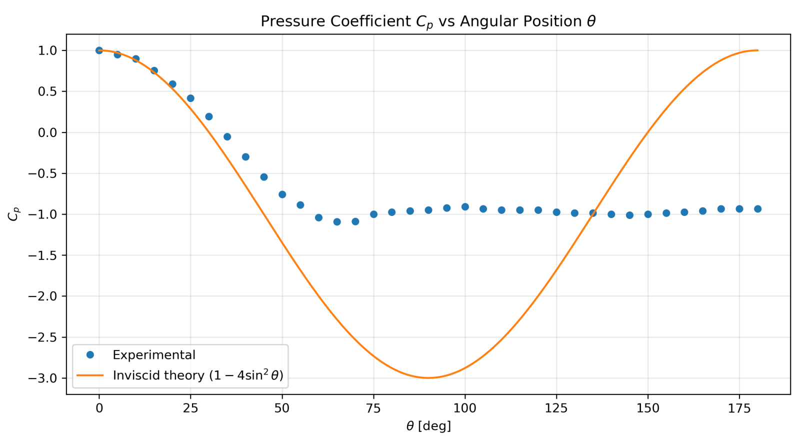

Figure 4.3.1: Pressure Coefficient C_p vs. Angle of Attack θ

4.3.3 Coefficient of Drag

The theoretically predicted drag coefficient for a cylinder in inviscid flow is zero. This result arises because, under the assumption of inviscid (frictionless) flow, the skin friction coefficient c_f is zero, and the symmetry of both the cylinder and the flow ensures that the pressure distribution is identical on the front and back surfaces. As a result, the opposing pressure forces cancel each other, leading to no net drag. This phenomenon is known as **D'Alembert's Paradox**.
Performing numerical analysis using the equations from the lecture:

$$C_d = \frac{D}{\rho_\infty} \tag{4.3.7}$$

$$D = R\int_0^{2\pi} P\cos(\theta)\,d\theta \tag{4.3.8}$$

Due to cylinder symmetry:

$$D = R\int_0^{2\pi} P\cos(\theta)\,d\theta = 2R\int_0^{\pi} P\cos(\theta)\,d\theta \tag{4.3.9}$$

To numerically calculate the integral, we used the trapezoidal approximation, and the results of C_d vs. θ are shown in the figure below:

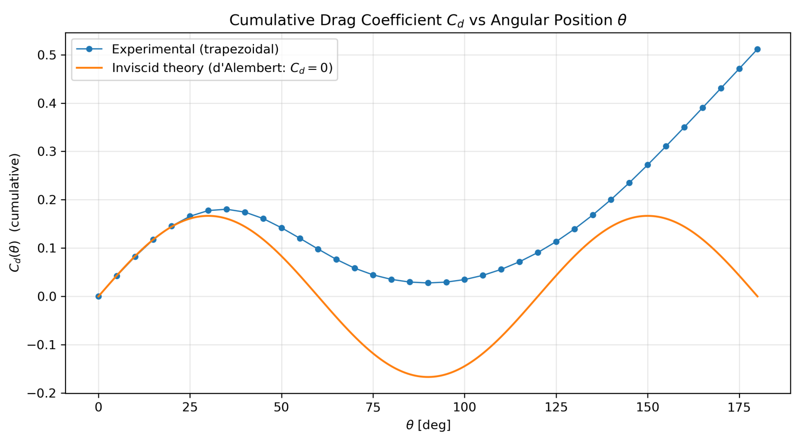

Figure 4.3.2: Drag Coefficient C_d vs. Angle of Attack θ

5 Analysis and Discussion

5.1 Data Conversion

The water manometer reports each pressure tap as a column height h on an attached ruler whose values increase downward. For any tap connected to the manometer, the actual pressure is related to the reading by

$$P_{\text{ref}} = P_{\text{tap}} - \rho_w\, g\, h, \tag{5.1.1}$$

where P_{ref} is a common reference pressure on the other side of the manometer and ρ_w is the density of water. Because P_{ref} is shared by all taps, it cancels when computing any differential pressure. In particular, the gauge pressure at the cylinder surface relative to the freestream is

$$P - P_\infty = \rho_w\, g\,(h_P - h_\infty)\,, \tag{5.1.2}$$

where h_∞ is the freestream static tap reading and h_P is the surface tap reading. A positive value indicates the surface pressure exceeds the freestream pressure, while a negative value indicates suction.

5.2 Interpretation

With regard to the coefficient of pressure, we see in Fig. 4.3.1 that distribution follows the inviscid prediction closely over the front portion of the cylinder (roughly $0^\circ \leq \theta \leq 30^\circ$), where both curves show a decrease from $C_p \approx 1$ at the stagnation point. This agreement is expected: the boundary layer on the forward face is thin and the flow behaves nearly as potential flow theory predicts.
Beyond about $\theta \approx 30^\circ - 40^\circ$, the experimental values begin to fall less steeply than the inviscid curve and diverge significantly. The inviscid theory reaches a minimum of $C_p =$ -3 at $\theta = 90^\circ$, whereas the experimental data levels off near $C_p \approx -1$ starting around $\theta \approx 70^\circ$ and remains essentially flat through the entire rear of the cylinder ($70^\circ - 180^\circ$). This flat, roughly constant C_p in the wake region is the hallmark of boundary layer separation.
As well, as can be seen in Fig. 4.3.2, the actual coefficient of drag is not zero. This is due to the fact that in real world air is viscous and it's coefficient of friction $c_f \neq 0$. This combined with potential flow detachment leads to a non-zero drag coefficient, for the same reason as the above coefficient of pressure discrepancies.

6 Summary and Conclusions

This experiment measured the surface pressure distribution on a circular cylinder in a vertical wind tunnel operating at a mean freestream velocity of 35.79 m/s. From the measurements, the pressure coefficient and drag coefficient were computed and compared against inviscid potential flow theory.
Three principal findings emerged from the analysis. First, the inviscid model accurately predicted the pressure distribution over the forward face of the cylinder, from the stagnation point at $\theta = 0^\circ$ to approximately $\theta = 30^\circ$, confirming that the flow in this region is wellapproximated by potential flow theory. Second, beyond roughly $\theta = 70^\circ$, the experimental pressure coefficient leveled off near $C_p \approx -1$ and remained essentially constant through the rear of the cylinder, rather than recovering symmetrically as predicted by inviscid theory. This behavior is characteristic of laminar boundary layer separation in the subcritical Reynolds number regime and demonstrates the limits of the inviscid flow assumption. Third, the pressure asymmetry between the front and rear of the cylinder produced a measurable net drag force, directly illustrating d'Alembert's paradox-the discrepancy between the zero-drag prediction of inviscid theory and the nonzero drag observed in viscous flow.
These results underscore that while potential flow theory provides a useful analytical framework for the pressure distribution on the upstream face of a bluff body, it cannot capture the effects of viscous separation that dominate the downstream flow field and are ultimately responsible for pressure drag.

7 References

[1] J. D. Anderson and C. P. Cadou, *Fundamentals of Aerodynamics*, 7th ed. McGraw-Hill Education, March 17, 2023, pp. 209–324, ISBN: 9781266076442 (see p. 1).

8 Appendices

8.1 Data

Table 8.1.1: Raw Measurements of Pressure Taps vs. Angle of Attack

`./data/pressure_vs_theta.csv`

	θ [°]	P_∞ [cm]	P_0 [cm]	P [cm]
01	0	15.4	23.0	23.0
02	5	15.4	23.1	22.7
03	10	15.4	23.1	22.3
04	15	15.4	23.2	21.3
05	20	15.3	23.1	19.9
06	25	15.4	23.1	18.6
07	30	15.4	23.2	16.9
08	35	15.4	23.1	15.0
09	40	15.4	23.1	13.1
10	45	15.4	23.1	11.2
11	50	15.3	23.1	9.4
12	55	15.3	23.1	8.4
13	60	15.4	23.1	7.4
14	65	15.4	23.1	7.0
15	70	15.4	23.2	6.9
16	75	15.4	23.2	7.6
17	80	15.4	23.2	7.8
18	85	15.4	23.1	8.0
19	90	15.4	23.1	8.1
20	95	15.4	23.1	8.3
21	100	15.4	23.2	8.3
22	105	15.4	23.1	8.2
23	110	15.4	23.1	8.1
24	115	15.4	23.1	8.1
25	120	15.4	23.1	8.1
26	125	15.4	23.1	7.9
27	130	15.4	23.1	7.8
28	135	15.4	23.1	7.8
29	140	15.4	23.1	7.7
30	145	15.4	23.1	7.6
31	150	15.4	23.1	7.7
32	155	15.4	23.1	7.8
33	160	15.4	23.0	8.0
34	165	15.4	23.1	8.0
35	170	15.4	23.1	8.2
36	175	15.4	23.1	8.2
37	180	15.4	23.1	8.2

8.2 Code

For the sake of brevity, only the code files that are key to the analysis are included below. However, in the spirit of completeness, the repository containing the complete data, source code, and notes for this report can be found at github:vaisriv/enae464-lab01.

Listing 8.2.1: Index File
`./src/index.py`

```python
import os
import sys
import csv
import numpy as np
import matplotlib.pyplot as plt

def main():
    # ── Physical constants & ambient conditions ─────────────────────────────
    P_AMB     = 100900.0 # Ambient pressure [Pa]
    T_AMB     = 298.15   # Ambient temperature [K]
    RHO_AIR   = 1.18     # Air density [kg/m^3]
    RHO_WATER = 998.0    # Water density [kg/m^3]
    G         = 9.81     # Gravitational acceleration [m/s^2]

    # ── File paths ──────────────────────────────────────────────────────────
    INPUT_CSV  = "./data/pressure_vs_theta.csv"
    OUTPUT_CSV = "./outputs/text/pressure_vs_theta.csv"
    OUTPUT_FILE = "./outputs/text/summary.txt"

    # ── Ensure figures output directory exists ─────────────────────────────
    FIG_DIR = os.path.join("outputs", "figures")
    os.makedirs(FIG_DIR, exist_ok=True)

    # ── Read CSV ────────────────────────────────────────────────────────────
    theta_deg_list = []
    P_inf_raw_list = []
    P_0_raw_list   = []
    P_raw_list     = []

    with open(INPUT_CSV, newline="") as f:
        reader = csv.DictReader(f)
        for row in reader:
            theta_deg_list.append(float(row["theta_deg"]))
            P_inf_raw_list.append(float(row["P_inf"]))
            P_0_raw_list.append(float(row["P_0"]))
            P_raw_list.append(float(row["P"]))

    N = len(theta_deg_list)
    print(f"Read {N} data points from '{INPUT_CSV}'.")

    # ── Convert water column heights to differential pressures ──────────────
```

```
    # The manometer readings are heights (in cm of water):
    #     ΔP = ρ_water · g · Δh
    #     P_actual = P_ref - ρ_water · g · h_reading
    #     P_surface - P_freestream  =  ρ_water · g · (h_surface - h_inf)
    #     q_inf = P_0 - P_inf = ρ_water · g · (h_0 - h_inf)
    SCALE = 1e-2  # Measurements taken in cm (convert to meters)

    # ── Compute differential pressures ──────────────────────────────────
    # For each data point we have a corresponding P_inf and P_0 reading,
    # so we compute per-row to account for any drift.
    delta_P_list = [] # P_surface - P_freestream  [Pa]
    q_inf_list   = [] # dynamic pressure  [Pa]
    U_inf_list   = [] # freestream velocity [m/s]
    Cp_list      = [] # pressure coefficient

    for i in range(N):
        h_inf = P_inf_raw_list[i] * SCALE # freestream static tap height [m]
        h_0   = P_0_raw_list[i]   * SCALE # stagnation tap height [m]
        h_p   = P_raw_list[i]     * SCALE # surface tap height [m]

        # (P_surface - P_inf) in Pa
        # Higher reading = higher pressure, so:
        delta_P = RHO_WATER * G * (h_p - h_inf)

        # Dynamic pressure  q = P_total - P_static = rho_w * g * (h_0 - h_inf)
        q_inf = RHO_WATER * G * (h_0 - h_inf)

        if q_inf ≤ 0:
            print(f"WARNING row {i}: q_inf = {q_inf:.2f} Pa ≤ 0  (h_inf={h_inf:.4f},
            ↪  h_0={h_0:.4f})")
            # Use absolute value as fallback but flag it
            q_inf = abs(q_inf) if abs(q_inf) > 1e-6 else 1e-6

        U_inf = (2.0 * q_inf / RHO_AIR) ** 0.5 # Bernoulli: q = 0.5 * rho * U^2

        Cp = delta_P / q_inf # Cp = (P - P_inf) / q_inf  = (P - P_inf) / (0.5 * rho * U^2)

        delta_P_list.append(delta_P)
        q_inf_list.append(q_inf)
        U_inf_list.append(U_inf)
        Cp_list.append(Cp)

    # ── Convert lists to numpy arrays for convenience ──────────────────
    theta_deg = np.array(theta_deg_list)
    theta_rad = np.deg2rad(theta_deg)
    Cp_exp    = np.array(Cp_list)
    dP_exp    = np.array(delta_P_list)
    q_inf_arr = np.array(q_inf_list)
```

```python

    # ── Inviscid (potential flow) theory for a cylinder ──────────────────────
    # Cp_inviscid = 1 - 4 sin^2(theta)
    theta_theory = np.linspace(0, 180, 500)
    theta_theory_rad = np.deg2rad(theta_theory)
    Cp_inviscid = 1.0 - 4.0 * np.sin(theta_theory_rad) ** 2

    # ── Figure 1: Cp vs theta ─────────────────────────────────────────────────
    fig1, ax1 = plt.subplots(figsize=(9, 5))
    ax1.plot(theta_deg, Cp_exp, 'o', markersize=5, label='Experimental')
    ax1.plot(theta_theory, Cp_inviscid, '-', linewidth=1.5, label='Inviscid theory ($1 - 4\\sin^2\\theta$)')
    ax1.set_xlabel(r'$\theta$ [deg]')
    ax1.set_ylabel(r'$C_p$')
    ax1.set_title(r'Pressure Coefficient $C_p$ vs Angular Position $\theta$')
    ax1.legend()
    ax1.grid(True, alpha=0.3)
    fig1.tight_layout()
    fig1.savefig(os.path.join(FIG_DIR, "Cp_vs_theta.png"), dpi=300)
    print(f"Saved {os.path.join(FIG_DIR, 'Cp_vs_theta.png')}")

    # ── Figure 2: Cd vs theta (cumulative drag coefficient) ──────────────────
    #
    # The drag force on the cylinder (per unit span) is:
    #     D = R ∫₀²π P·cos(θ) dθ
    #
    # We define the drag coefficient as:
    #     C_d = D / (q_inf · d)  =  D / (q_inf · 2R)
    #
    # Substituting and non-dimensionalising with Cp = (P - P_inf) / q_inf :
    #     C_d = (1/2) ∫₀²π Cp · cos(θ) dθ
    #
    # (The P_inf contribution integrates to zero over a closed surface.)
    #
    # We approximate the integral cumulatively using the trapezoidal rule so we
    # can plot C_d as a function of the upper integration limit θ.

    # Sort by angle to ensure proper integration order
    sort_idx  = np.argsort(theta_deg)
    theta_sorted     = theta_deg[sort_idx]
    theta_sorted_rad = theta_rad[sort_idx]
    Cp_sorted        = Cp_exp[sort_idx]

    # Integrand: Cp(θ) · cos(θ)
    integrand_exp = Cp_sorted * np.cos(theta_sorted_rad)

    # Cumulative trapezoidal integration: (1/2) ∫₀^θ Cp·cos(θ') dθ'
    Cd_cumulative_exp = np.zeros(len(theta_sorted))
```

```
    for j in range(1, len(theta_sorted)):
        dtheta = theta_sorted_rad[j] - theta_sorted_rad[j - 1]
        Cd_cumulative_exp[j] = Cd_cumulative_exp[j - 1] + 0.5 * (integrand_exp[j] + integrand_exp[j - 1]) * dtheta
    Cd_cumulative_exp *= 0.5  # the 1/2 prefactor from C_d = (1/2) ∫ Cp cos(θ) dθ

    # Inviscid theory: Cp_inv = 1 - 4sin²θ  →  Cd = 0 (d'Alembert's paradox)
    # Cumulative for plotting:
    integrand_inv = (1.0 - 4.0 * np.sin(theta_theory_rad) ** 2) * np.cos(theta_theory_rad)
    Cd_cumulative_inv = np.zeros(len(theta_theory))
    for j in range(1, len(theta_theory)):
        dtheta = theta_theory_rad[j] - theta_theory_rad[j - 1]
        Cd_cumulative_inv[j] = Cd_cumulative_inv[j - 1] + 0.5 * (integrand_inv[j] + integrand_inv[j - 1]) * dtheta
    Cd_cumulative_inv *= 0.5

    fig2, ax2 = plt.subplots(figsize=(9, 5))
    ax2.plot(theta_sorted, Cd_cumulative_exp, 'o-', markersize=4, linewidth=1, label='Experimental (trapezoidal)')
    ax2.plot(theta_theory, Cd_cumulative_inv, '-', linewidth=1.5, label="Inviscid theory (d'Alembert: $C_d = 0$)")
    ax2.set_xlabel(r'$\theta$ [deg]')
    ax2.set_ylabel(r'$C_d(\theta)$  (cumulative)')
    ax2.set_title(r'Cumulative Drag Coefficient $C_d$ vs Angular Position $\theta$')
    ax2.legend()
    ax2.grid(True, alpha=0.3)
    fig2.tight_layout()
    fig2.savefig(os.path.join(FIG_DIR, "Cd_vs_theta.png"), dpi=300)
    print(f"Saved {os.path.join(FIG_DIR, 'Cd_vs_theta.png')}")

    # Print final Cd value (full integral over 0 to max angle)
    print(f"\nExperimental C_d (integrated 0 to {theta_sorted[-1]:.0f} deg): {Cd_cumulative_exp[-1]:.4f}")
    print(f"Inviscid theory C_d (integrated 0 to 360 deg):      {Cd_cumulative_inv[-1]:.6f} (≈ 0, d'Alembert's paradox)")

    # ── Summary statistics ──────────────────────────────────────────
    q_avg   = sum(q_inf_list) / N
    U_avg   = sum(U_inf_list) / N

    with open(OUTPUT_FILE, "w", newline="") as f:
        f.write(f"Ambient conditions:")
        f.write(f"  P_amb   = {P_AMB} Pa")
        f.write(f"  T_amb   = {T_AMB} K")
        f.write(f"  rho_air = {RHO_AIR} kg/m^3")
        f.write(f"\nDerived quantities (averages over all rows):")
        f.write(f"  q_inf   = {q_avg:.2f} Pa")
        f.write(f"  U_inf   = {U_avg:.2f} m/s")
```

```python

    print(f"\nSummary written to '{OUTPUT_FILE}'.")

    # ── Write output table ──────────────────────────────────────────
    with open(OUTPUT_CSV, "w", newline="") as f:
        writer = csv.writer(f)
        writer.writerow(["theta_deg", "P-P_inf_Pa", "Cp", "P_Pa"])

        for i in range(N):
            # P_actual = P_amb + (P_surface - P_inf) i.e. ambient + differential
            P_actual = P_AMB + delta_P_list[i]
            writer.writerow([
                f"{theta_deg_list[i]:.1f}",
                f"{delta_P_list[i]:.3f}",
                f"{Cp_list[i]:.4f}",
                f"{P_actual:.3f}",
            ])

    print(f"\nResults written to '{OUTPUT_CSV}'.")


if __name__ == "__main__":
    main()
```